

# A Disproof of Conjecture 00000001262

The abelian complexity of a balanced periodic word is not eventually constant: the word  $(aab)^\infty$

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## Abstract

Conjecture 00000001262 states: *the abelian complexity of balanced periodic words is eventually constant, equal to the alphabet size (the abelian constant class)*. We disprove it with the word  $w = (aab)^\infty$ , which is balanced and periodic over the alphabet  $\{a, b\}$  of size 2, yet whose abelian complexity satisfies  $\omega(3k) = 1$  for *every*  $k \geq 0$ : it equals 1 at the infinitely many lengths  $n = 3, 6, 9, \dots$  and therefore is not eventually constant, let alone eventually equal to 2. The full universal statement  $\forall k, \omega(3k) = 1$  and the limit negation  $\neg \exists N \forall n \geq N, \omega(n) = 2$  are machine-checked in Lean 4 (core library only, zero axioms, zero `sorry`).

## 1 Setup: abelian complexity

Let  $w$  be a one-sided infinite word over a finite alphabet. For  $n \geq 1$  a *factor* (window) of length  $n$  is  $w[s..s+n-1]$ , a contiguous block starting at position  $s$ . Its *Parikh vector* records how many times each letter occurs. The *abelian complexity* of  $w$  is

$$\omega(n) = \#\{\text{Parikh vectors of length-}n \text{ windows of } w\}.$$

Since the Parikh vector of a length- $n$  window is determined by its number  $A(s, n)$  of letters  $a$  (the number of  $b$ 's is  $n - A(s, n)$ ),  $\omega(n)$  is the number of *distinct values* of  $A(s, n)$  over window starts  $s$ .

## 2 The counterexample word and its two structural properties

Let

$$w = (aab)^\infty = aabaaabaaab \dots$$

over  $\{a, b\}$ , an alphabet of size  $q = 2$ .

**Lemma 1** (Periodicity).  *$w$  is periodic with period 3: the letter at position  $i+3$  equals the letter at position  $i$  for all  $i$ .*

*Proof.* Position  $i$  carries  $b$  iff  $i \equiv 2 \pmod{3}$ , and  $i+3 \equiv i \pmod{3}$ . □

**Lemma 2** (Balance).  *$w$  is balanced: any two windows of the same length contain numbers of  $a$ 's differing by at most 1.*

*Proof.* Shifting a window by one position removes one letter and appends one letter, so  $|A(s+1, n) - A(s, n)| \leq 1$ . By Lemma 1,  $A(s+3, n) = A(s, n)$  (a shift by 3 is a cyclic permutation of the period). Hence as  $s$  ranges over all starts,  $A(s, n)$  takes at most the three values  $A(0, n), A(1, n), A(2, n)$ , and consecutive ones among these differ by at most 1; so the range of  $A(\cdot, n)$  has diameter at most 1. □

Thus  $w$  is a *balanced periodic word* in the sense of the conjecture.

### 3 The exact abelian complexity of $(aab)^\infty$

Write  $A_s(n)$  for the number of  $a$ 's in the window starting at  $s$  of length  $n$ .

**Lemma 3** (Shift by one).  $A_s(n+1) = \chi(s) + A_{s+1}(n)$ , where  $\chi(s) = 1$  if position  $s$  carries  $a$  and  $\chi(s) = 0$  if it carries  $b$ ; explicitly  $\chi(0) = \chi(1) = \chi(2) = \chi(3) = \chi(4) = 1$ ,  $\chi(5) = 0$ .

**Lemma 4** (Length  $3k$ ). For every  $k \geq 0$  and every start  $s$ :  $A_s(3k) = 2k$ .

*Proof.* By induction on  $k$ . For  $k = 0$  the window is empty. For the step, unwrap three letters (Lemma 3 three times):

$$A_s(3k+3) = \chi(s) + \chi(s+1) + \chi(s+2) + A_{s+3}(3k).$$

Every three consecutive positions contain exactly two  $a$ 's (the window  $(s, s+1, s+2)$  is a cyclic permutation of  $aab$ ), and  $A_{s+3}(3k) = A_s(3k)$  by periodicity (Lemma 1), so  $A_s(3k+3) = 2 + A_s(3k) = 2k + 2$ .  $\square$

**Proposition 1.** For all  $k \geq 0$ :

$$\omega(3k) = 1, \quad \omega(3k+1) = 2, \quad \omega(3k+2) = 2.$$

*Proof.* By Lemma 4,  $A_0(3k) = A_1(3k) = A_2(3k) = 2k$ ; since these three starts cover all Parikh classes (Lemma 2 argument), every length- $3k$  window has Parikh vector  $(2k, k)$ , whence  $\omega(3k) = 1$ . For  $3k+1$ : unwrapping one letter gives  $A_0(3k+1) = \chi(0) + A_1(3k) = 2k+1$ ,  $A_1(3k+1) = \chi(1) + A_2(3k) = 2k+1$ ,  $A_2(3k+1) = \chi(2) + A_0(3k) = 2k$ , so exactly two values occur:  $\omega(3k+1) = 2$ . For  $3k+2$ :  $A_0(3k+2) = 2k+2$  and  $A_1(3k+2) = A_2(3k+2) = 2k+1$ , so  $\omega(3k+2) = 2$ .  $\square$

The first values, computable by hand:

| $n$         | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|-------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|
| $\omega(n)$ | 2 | 2 | 1 | 2 | 2 | 1 | 2 | 2 | 1 | 2  | 2  | 1  | 2  | 2  |

### 4 Refutation of the conjecture

The conjecture predicts that  $\omega(n)$  is eventually constant with value equal to the alphabet size, here 2.

**Proposition 2** (Limit negation).  $\neg \exists N \forall n \geq N, \omega(n) = 2$ .

*Proof.* Given any  $N$ , put  $n = 3(N+1) \geq N$ . By Proposition 1,  $\omega(n) = 1 \neq 2$ . So no such  $N$  exists.  $\square$

**Remark 1** (Why the conjecture fails here, precisely). At lengths  $n \not\equiv 0 \pmod{3}$  the word does exhibit the conjectured value  $\omega(n) = 2$ ; the letter multiplicities just fail to stabilize because at every multiple of 3 the factorization into whole periods collapses the possible Parikh vectors to the single vector  $(2k, k)$ . Balance constrains  $\omega(n) \leq 2$  for this word but cannot prevent the periodic collapse to 1.

**Remark 2** (Scope). The conjecture is a universal statement over all balanced periodic words; one counterexample refutes it. No finite enumeration stands in for the infinite statement: the formalization proves  $\forall k, \omega(3k) = 1$  as a quantified theorem, together with the limit negation  $\neg \exists N \forall n \geq N, \omega(n) = 2$ , and proves  $\omega(3k+1) = \omega(3k+2) = 2$  to show the word meets the conjectured value at the other residues.

## 5 Formal verification

The Lean development (`lean4/`, toolchain `leanprover/lean4:v4.33.1`, core library only) formalizes everything above in namespace `Tlmc1262`:

- `omega_triple` :  $\forall k, \text{omega } (3*k) = 1$ ;
- `omega_triple_succ`, `omega_triple_succ2`: the = 2 residues;
- `not_eventually_two` :  $\neg \exists N, \forall n \geq N, \text{omega } n = 2$ ;
- faithfulness of the three-start definition of  $\omega$ : `A_shiftQ` :  $\forall q \ r \ n, A \ (3*q + r) \ n = A \ r \ n$ .

Because the core library's lemmas about `%` (e.g. `Nat.add_mod_left`) depend on `propext`, the 3-periodicity of the letter function is derived from scratch via an explicit three-phase clock built by structural recursion (`cycle`, `ph`). `Check.lean` audits all 27 theorems with `#print axioms` and with a metaprogram that throws on any axiom use: all are **axiom-free** (no `propext`, no `Classical.choice`, no `Quot.sound`, no `sorryAx`). `reproduce.py` independently recomputes the  $\omega$  table, balance, and the limit negation.